

Gagan Bansal

Chandigarh, India — bansalgagan2004@gmail.com — +91 9996892070
LinkedIn — GitHub — LeetCode — Portfolio

Education

Chitkara University, Chandigarh
B.E Computer Science

Oct 2022 – June 2026

CGPA: 9.41

Allen Career Institute, Kota
Senior Secondary (Non-Medical)

Apr 2020 – Jun 2022

Technical Skills

Languages: C++, Java, C, JavaScript

Systems: Multithreading, Concurrency, TCP Networking, Memory Management

Web and Backend: React.js, Node.js, Express.js, Apache Kafka, REST APIs, SOAP

Databases: MongoDB, SQL, PostgreSQL

Tools and Platforms: Git, Linux, macOS, AWS (Basic)

Core CS: Data Structures and Algorithms, OOP, Operating Systems, Computer Networks, DBMS

Experience

Research and Development Intern, *Quark Software Inc.*

Jan 2026 – Present *Chandigarh*

- Porting and modernizing a legacy codebase to ensure compatibility with the latest macOS versions, supporting both ARM64 (Apple Silicon) and x86-64 architectures.
- Resolved complex cross-platform build and dependency issues involving OpenSSL and Axis2, ensuring stable builds across ARM64 and x86 architectures.
- Debugging and fixing low-level compatibility issues arising from architecture differences and deprecated system APIs.

Projects

KVMemo – Multithreaded In-Memory Key-Value Store – C++

kvmemo.dev — GitHub

- Designed and implemented a high-performance in-memory key-value store in modern C++ supporting concurrent TCP client connections.
- Built a thread-pool architecture with task scheduling and synchronization using `std::mutex`, `std::sharedmutex`, and atomic counters to optimize read-heavy workloads.
- Implemented runtime metrics including QPS, latency tracking, hit/miss ratio, and active connection monitoring for performance observability.

oForge – Engineering Workspace Platform

GitHub

- Built oForge as a team collaboration platform for students, combining team discovery with a feature-rich workspace including Kanban boards, real-time chat, schema design, documentation, and whiteboard tools.
- Introduced a gated team collaboration system where users must submit proposals to join projects, with workspace access granted only after owner approval, ensuring privacy and controlled team formation.

CDS Library – C

GitHub

- Built modular header files for String, Stack, and Linked List with reusable C interfaces.
- Improved memory management and reduced redundancy using macros and pointer abstraction.

Achievements

- Solved 1300+ Data Structures and Algorithms problems across LeetCode, GFG, and CodeChef.
- Achieved LeetCode Contest Rating of **1618**.
- Selected for UCA Batch at Chitkara University – Top 3% out of 2500+ students.
- **Finalist** – Shopclues Smart Cataloging Hackathon (Top 5 of 70+ teams), Chandigarh, 2024.